Lab Sheet-02

BCA 3rd Semester (Session 2026-2027)

1. Student Registration Form: Create a Student Registration Form using HTML. Include text fields, radio buttons, checkboxes, dropdown list, textarea, submit, and reset buttons. Apply CSS to style the form.

Ans.

```
<html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Student Registration</title>
  <style>
    body { font-family: Arial, sans-serif; background: #f1f5f9; }
    form {
      max-width: 500px; margin: 30px auto; padding: 24px;
      background: white; border: 1px solid #cbd5e1; border-radius: 10px;
    }
    h1 { text-align: center; color: #1d4ed8; }
    label { display: block; margin-top: 14px; }
    input:not([type="radio"]):not([type="checkbox"]), select, textarea {
      box-sizing: border-box; width: 100%; padding: 9px; margin-top: 5px;
    }
    .options label { display: inline-block; margin-right: 12px; }
    button { padding: 10px 16px; margin: 18px 8px 0 0; border: 0; cursor: pointer; }
    .submit { background: #2563eb; color: white; }
  </style>
</head>
<body>
  <form>
    <h1>Student Registration</h1>
```

```
    <label for="name">Full Name</label>
    <input id="name" name="name" type="text" required>

    <label for="email">Email</label>
    <input id="email" name="email" type="email" required>

    <label for="course">Course</label>
    <select id="course" name="course">
      <option value="">Choose a course</option>
      <option>Computer Science</option>
      <option>Business</option>
      <option>Arts</option>
    </select>

    <label>Gender</label>
    <div class="options">
      <label><input type="radio" name="gender" value="female"> Female</label>
      <label><input type="radio" name="gender" value="male"> Male</label>
      <label><input type="radio" name="gender" value="other"> Other</label>
    </div>

    <label>Interests</label>
    <div class="options">
      <label><input type="checkbox" name="interest" value="sports">
Sports</label>
      <label><input type="checkbox" name="interest" value="music">
Music</label>
      <label><input type="checkbox" name="interest" value="coding">
Coding</label>
    </div>

    <label for="address">Address</label>
    <textarea id="address" name="address" rows="4"></textarea>

    <button class="submit" type="submit">Submit</button>
    <button type="reset">Reset</button>
  </form>
</body>
</html>
```

2 College Website using Frames: Create a college website using HTML Frames with Header, Navigation Menu, and Main Content sections. Add navigation links that display different pages in the main frame.

Ans.

```
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>COER College</title>
  <style>
    * { box-sizing: border-box; }
    body {
      margin: 0;
      font-family: Arial, sans-serif;
      display: grid;
      grid-template-rows: 100px 1fr;
      height: 100vh;
    }
    header {
      background: #1e3a8a;
      color: white;
      text-align: center;
      padding: 20px;
    }
    .layout {
      display: grid;
      grid-template-columns: 210px 1fr;
      min-height: 0;
    }
    nav {
      background: #e2e8f0;
      padding: 20px;
    }
    nav a {
      display: block;
      color: #1e3a8a;
      padding: 12px;
      margin-bottom: 6px;
      text-decoration: none;
      border-radius: 5px;
    }
    nav a:hover { background: #cbd5e1; }
```

```
    main { min-width: 0; }
    iframe { width: 100%; height: 100%; border: 0; }
  </style>
</head>
<body>
  <header>
    <h1>COER UNIVERSITY </h1>
    <p>Learn, grow, and shape your future</p>
  </header>

  <div class="layout">
    <nav aria-label="Main navigation">
      <h2>Menu</h2>
      <a href="#home" onclick="showPage('home')">Home</a>
      <a href="#courses" onclick="showPage('courses')">Courses</a>
      <a href="#admission" onclick="showPage('admission')">Admissions</a>
      <a href="#contact" onclick="showPage('contact')">Contact</a>
    </nav>

    <main>
      <iframe id="contentFrame" title="College information"></iframe>
    </main>
  </div>

  <script>
    const pages = {
      home: `
        <h2>Welcome to COER UNIVERSITY</h2>
        <p>We help students build knowledge, skills, and confidence for the
future.</p>
      `,
      courses: `
        <h2>Our Courses</h2>
        <ul>
          <li>Computer Science</li>
          <li>Business Studies</li>
          <li>Arts and Humanities</li>
        </ul>
      `,
      admission: `
        <h2>Admissions</h2>
        <p>Applications are open to students interested in joining our
programs.</p>
```

```
      <p>Contact the admissions office for application details.</p>
    `,
    contact: `
      <h2>Contact Us</h2>
      <p>Email: coer@gmail.edu</p>
      <p>Phone: 123-456-7890</p>
    `
  };

  function showPage(page) {
    const frame = document.getElementById("contentFrame");
    const content = pages[page] || pages.home;
    const frameDocument = frame.contentDocument;

    frameDocument.open();
    frameDocument.write(`
      <!DOCTYPE html>
      <html lang="en">
      <head>
        <meta charset="UTF-8">
        <style>
          body { font-family: Arial, sans-serif; padding: 24px; color: #1f2937; }
          h2 { color: #1e3a8a; }
        </style>
      </head>
      <body>${content}</body>
      </html>
    `);
    frameDocument.close();
  }

  showPage("home");
 </script>
</body>
</html>
```

3 Login and Admission Form: Design a Login Form and College Admission Form using HTML. Use appropriate form elements and apply CSS for formatting, borders, spacing, fonts, and buttons.

Ans.

```
 <html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>College Forms</title>
  <style>
    body { font-family: Arial, sans-serif; background: #f1f5f9; padding: 20px; }
    form {
      max-width: 440px; margin: 20px auto; padding: 22px;
      background: white; border: 1px solid #cbd5e1; border-radius: 8px;
    }
    h2 { color: #1d4ed8; }
    label { display: block; margin-top: 12px; }
    input, select, textarea {
      box-sizing: border-box; width: 100%; padding: 9px; margin-top: 5px;
      border: 1px solid #94a3b8; border-radius: 4px;
    }
    button { margin-top: 16px; padding: 10px 16px; background: #2563eb;
            color: white; border: 0; border-radius: 4px; cursor: pointer; }
  </style>
</head>
<body>
  <form>
    <h2>Login</h2>
    <label for="login-email">Email</label>
    <input id="login-email" type="email" required>
    <label for="login-password">Password</label>
    <input id="login-password" type="password" required>
    <button type="submit">Log In</button>
  </form>

  <form>
    <h2>College Admission</h2>
    <label for="student-name">Full Name</label>
    <input id="student-name" type="text" required>
    <label for="admission-email">Email</label>
    <input id="admission-email" type="email" required>
    <label for="phone">Phone Number</label>
```

```
    <input id="phone" type="tel" required>
    <label for="program">Program</label>
    <select id="program" required>
      <option value="">Select a program</option>
      <option>Computer Science</option>
      <option>Business Studies</option>
      <option>Arts</option>
    </select>
    <label for="details">Previous Education</label>
    <textarea id="details" rows="3"></textarea>
    <button type="submit">Apply</button>
  </form>
</body>
</html>
```

4. JavaScript Calculator: Create a simple calculator using HTML and JavaScript to perform addition, subtraction, multiplication, and division of two numbers.

Ans.

```
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Calculator</title>

  <style>
    * { box-sizing: border-box; }

    body {
      margin: 0;
      min-height: 100vh;
      display: grid;
      place-items: center;
      background: #10131c;
      font-family: Arial, sans-serif;
    }

    .calculator {
      width: min(360px, 92vw);
      padding: 24px;
      background: #1d2330;
      border: 1px solid #343d4f;
      border-radius: 20px;
      box-shadow: 0 18px 50px #0008;
    }

    h1 {
      margin: 0 0 20px;
      color: #f4f6fb;
      text-align: center;
      font-size: 24px;
    }

    .display {
      min-height: 72px;
      margin-bottom: 18px;
      padding: 20px 14px;
      overflow-wrap: anywhere;
```

```
  border-radius: 12px;
  background: #111722;
  color: #8df0c6;
  text-align: right;
  font-size: 26px;
}

.fields {
  display: grid;
  gap: 10px;
  margin-bottom: 16px;
}

input {
  width: 100%;
  padding: 13px;
  border: 1px solid #3b4659;
  border-radius: 10px;
  outline: none;
  background: #252d3b;
  color: white;
  font-size: 16px;
}

input:focus { border-color: #8df0c6; }

.buttons {
  display: grid;
  grid-template-columns: repeat(4, 1fr);
  gap: 10px;
}

button {
  padding: 15px 0;
  border: 0;
  border-radius: 10px;
  background: #30394a;
  color: white;
  font-size: 21px;
  cursor: pointer;
}

button:hover { background: #414d62; }
```

```
    button.operator {
      background: #186e58;
      color: #eafff7;
    }

    button.operator:hover { background: #21856b; }

    button.clear {
      grid-column: 1 / -1;
      background: #a6404c;
      font-size: 15px;
    }

    button.clear:hover { background: #c2505d; }
  </style>
</head>
<body>
  <main class="calculator">
    <h1>Quick Calculator</h1>

    <div class="display" id="answer" aria-live="polite">Ready</div>

    <div class="fields">
      <input id="firstNumber" type="number" step="any" placeholder="First
number">
      <input id="secondNumber" type="number" step="any" placeholder="Second
number">
    </div>

    <div class="buttons">
      <button class="operator" onclick="calculate('+')" aria-label="Add">+</button>
      <button class="operator" onclick="calculate('-')" aria-
label="Subtract">−</button>
      <button class="operator" onclick="calculate('*')" aria-
label="Multiply">×</button>
      <button class="operator" onclick="calculate('/')" aria-
label="Divide">÷</button>
      <button class="clear" onclick="clearCalculator()">Clear</button>
    </div>
  </main>

  <script>
```

```
    function calculate(operation) {
      const firstText = document.getElementById("firstNumber").value;
      const secondText = document.getElementById("secondNumber").value;
      const display = document.getElementById("answer");

      if (firstText === "" || secondText === "") {
        display.textContent = "Enter both numbers";
        return;
      }

      const first = Number(firstText);
      const second = Number(secondText);
      let result;

      switch (operation) {
        case "+":
          result = first + second;
          break;
        case "-":
          result = first - second;
          break;
        case "*":
          result = first * second;
          break;
        case "/":
          if (second === 0) {
            display.textContent = "Cannot divide by zero";
            return;
          }
          result = first / second;
          break;
      }

      display.textContent = "Answer: " + result;
    }
      function clearCalculator() {
      document.getElementById("firstNumber").value = "";
      document.getElementById("secondNumber").value = "";
      document.getElementById("answer").textContent = "Ready";
    }
  </script>
</body>
</html>
```

5 JavaScript Validation: Create a Student Registration Form using HTML and JavaScript. Validate fields such as Name, Email, Mobile Number, Password, and Age before submitting the form. Display appropriate validation messages.

Ans.

```
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Student Registration</title>

  <style>
    * { box-sizing: border-box; }

    body {
      margin: 0;
      min-height: 100vh;
      display: grid;
      place-items: center;
      padding: 24px;
      background: linear-gradient(135deg, #dbeafe, #f5e8ff);
      font-family: Arial, sans-serif;
    }

    .card {
      width: min(480px, 100%);
      padding: 30px;
      background: white;
      border-radius: 18px;
      box-shadow: 0 14px 40px #1e3a8a22;
    }

    h1 {
      margin: 0 0 8px;
      color: #23395d;
    }

    .intro {
      margin: 0 0 22px;
      color: #64748b;
    }
```

```
label {
  display: block;
  margin: 14px 0 6px;
  color: #334155;
  font-weight: bold;
}

input {
  width: 100%;
  padding: 12px;
  border: 1px solid #cbd5e1;
  border-radius: 8px;
  font-size: 15px;
}

input:focus {
  outline: 2px solid #93c5fd;
  border-color: #3b82f6;
}

.buttons {
  display: flex;
  gap: 10px;
  margin-top: 22px;
}

button {
  flex: 1;
  padding: 12px;
  border: 0;
  border-radius: 8px;
  font-size: 15px;
  font-weight: bold;
  cursor: pointer;
}

.submit {
  background: #2563eb;
  color: white;
}

.submit:hover { background: #1d4ed8; }
```

```
    .reset {
      background: #e2e8f0;
      color: #334155;
    }

    #message {
      margin: 16px 0 0;
      line-height: 1.6;
    }

    .error { color: #b91c1c; }
    .success { color: #15803d; }
  </style>
</head>
<body>
  <main class="card">
    <h1>Student Registration</h1>
    <p class="intro">Complete the form to register.</p>

    <form id="registrationForm" novalidate>
      <label for="name">First Name</label>
      <input id="name" type="text" placeholder="Enter your name">

      <label for="name">Last Name</label>
      <input id="name" type="text" placeholder="Enter your name">



      <label for="email">Email Address</label>
      <input id="email" type="text" placeholder="name@example.com">

      <label for="mobile">Mobile Number</label>
      <input id="mobile" type="text" placeholder="10-digit mobile number">

      <label for="password">Password</label>
      <input id="password" type="password" placeholder="At least 8 characters">

      <label for="age">Age</label>
      <input id="age" type="number" placeholder="Enter your age">

      <div class="buttons">
        <button class="submit" type="submit">Register</button>
        <button class="reset" type="reset">Clear Form</button>
```

```
    </div>

    <p id="message" aria-live="polite"></p>
  </form>
</main>

<script>
  const form = document.getElementById("registrationForm");
  const message = document.getElementById("message");

  form.addEventListener("submit", function(event) {
    event.preventDefault();

    const name = document.getElementById("name").value.trim();
    const email = document.getElementById("email").value.trim();
    const mobile = document.getElementById("mobile").value.trim();
    const password = document.getElementById("password").value;
    const ageText = document.getElementById("age").value;
    const age = Number(ageText);
    const errors = [];

    if (name.length < 2) {
      errors.push("Name must contain at least 2 characters.");
    }

    if (!/^[^\s@]+@[^\s@]+\.[^\s@]+$/.test(email)) {
      errors.push("Enter a valid email address.");
    }

    if (!/^\d{10}$/.test(mobile)) {
      errors.push("Mobile number must contain exactly 10 digits.");
    }

    if (password.length < 8) {
      errors.push("Password must contain at least 8 characters.");
    }

    if (ageText === "" || !Number.isInteger(age) || age < 16 || age > 100) {
      errors.push("Age must be a whole number between 16 and 100.");
    }

    if (errors.length > 0) {
      message.className = "error";
```

```
      message.innerHTML = errors.map(error => "• " + error).join("<br>");
    } else {
      message.className = "success";
      message.textContent = "Registration successful!";
    }
  });

  form.addEventListener("reset", function() {
    message.textContent = "";
    message.className = "";
  });
 </script>
</body>
</html>
```